Jianxuan Shao

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RESEARCH INTERESTS

Embodied and spatial interaction; context-aware AI assistance; user control in immersive environments.

Computer science master's student with a background in architecture and immersive media. I am interested in designing and evaluating XR and gesture-based interfaces that connect people's actions with digital and physical environments.

EDUCATION

University of Sydney Feb 2025 - Present

Master of Computer Science (Data Science and AI), expected Dec 2026

Average: 79.33/100 across 12 graded units; Semester 1, 2026: 84.5/100. Selected results: Introduction to Cybersecurity 92; Machine Learning and Data Mining 89; Introduction to Programming 87; Computer and Network Organisation 85.

Hebei University Sep 2018 - Jun 2023

Bachelor of Engineering in Architecture

GPA: 4.05/5.0. Five-year training in architectural design, architectural physics, urban planning, computational design and digital visualisation.

IMMERSIVE MEDIA AND SPATIAL DESIGN

Shanghai Fengyuzhu Culture Technology Co., Ltd. Jul 2023 - Jul 2024

Interactive Media Intern

- Supported immersive spatial systems integrating CAVE multi-channel projection, wall-touch interfaces, radar recognition and VR/AR visualisation.
- Worked across design, technical delivery and on-site system debugging to translate spatial concepts into user-facing interactive experiences.

Daci Pavilion and Geyuan Temple Reconstruction Oct 2021

Surveying and digital reconstruction | Team Member

- Recorded historic architectural geometry through sketches and CAD/Rhino documentation; reconstructed plans, elevations, sections and three-dimensional information for preservation.
- Modelled architectural components, including traditional bracket forms. Project records were archived in Baoding City Library.

INTERACTIVE PROTOTYPING AND STUDY PREPARATION

Game Fellows Initiated 2025

AI game companion | Prototype completed

- Developed a Python/Pygame co-play prototype by extending a Galaga-style game with a visible, rule-based teammate for threat interception, shielding and coordinated support fire.
- Linked text and audio callouts to support actions, with cooldowns to limit repeated prompts and rapid behaviour switching.
- Prepared a counterbalanced Solo/Fellow study with paired random seeds, gameplay/event logs and questionnaires on presence, trust and player control. No recruited-user study was conducted.

COMPUTATIONAL RESEARCH AND PROJECTS

Explainable AI Generated Image Detection 2026 - Present

University of Sydney capstone research project | Eight-student team

- Contributed across research framing, implementation, evaluation and error analysis to investigate how AI-generated image detectors can provide checkable visual evidence.
- Contributed to a framework combining Qwen3.5-9B detection-first/evidence-first reasoning, independent ResNet-50 visual forensics and logistic fusion on 6,000 FakeBench images from 10 generators.
- Team result: 94.33% balanced accuracy on a 900-image test set. Sample-level analysis found 45 of 51 fusion errors among 275 cross-system disagreement cases, suggesting a potential signal for human review.
- Explanation quality was assessed on 200 fake images using automatic metrics and a model judge. Generalisation to unseen generators and human-rated explanation quality remain untested.

Machine Learning Classification and PathMNIST 2026

COMP5318 coursework project

- Built Python/scikit-learn workflows with preprocessing, stratified cross-validation and hyperparameter tuning; implemented Random Forest, MLP and CNN baselines for PathMNIST image classification.
- Compared accuracy, macro-F1, confusion matrices and runtime trade-offs, with attention to class-level errors and model behaviour.

Electricity Data Engineering and Spatial Integration 2026

COMP5339 coursework project

- Integrated electricity, emissions, renewable-station and regional data using Python, geocoded coordinates and spatial DuckDB storage. Processed 716,454 facility records and streamed JSON via MQTT to a Streamlit map dashboard.

ADDITIONAL ARCHITECTURE EXPERIENCE

Hebei Haoyu Architectural Design Co., Ltd. Jun 2022 - Aug 2022

Beidaihe Branch | Architectural Design Intern

- Contributed to an experimental primary-school project through site and user-needs research, spatial layout studies and client feedback iterations; produced CAD plans, elevations, sections and visualisations.

Qinhuangdao Guangjian Construction Project Management Jun 2021 - Aug 2021

Intern / Assistant Architect

- Produced demolition, facade, floor-plan and sectional drawings for a shantytown redevelopment project under senior architects' design requirements.

TECHNICAL SKILLS AND TRAINING

Tools: Python, SQL, Pygame, scikit-learn, DuckDB, MQTT and Streamlit; CAD and Rhino for architectural documentation and 3D modelling.

Methods: Reproducible model evaluation, error analysis, controlled study preparation, interactive prototyping and spatial design.

Further training: Rhino/Grasshopper workshops; SCI-Arc digital visualisation workshop; UIA-HYP competitions (2021, 2022); Android authentication threat modelling and technical reporting (INFO5995).

AWARDS AND LEADERSHIP

Hebei University Outstanding Student Scholarship: Second Prize (2019); Third Prize (2021, 2022). JYZC Undergraduate Innovation and Creativity Contest: Nomination Award (2022).

President, Hebei University Tennis Club (2019-2020). Team Leader, Hebei University architecture-themed campus sketching activity (2022).